



ComicCraft

AI Comic Story Creator using Gemini Models



Team Members

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Project Overview



ComicCraft is a web-based application that uses AI to generate personalized comic book stories and illustrations based on user-provided prompts. It uses FastAPI and integrates Google's Gemini AI models along with Stable Diffusion to streamline the creative process of generating storylines, dialogues, and vivid comic-style imagery automatically. Users can view the comic preview on the web interface and download the full comic in PDF format.

- Built with FastAPI and Google's Gemini AI models
- Illustrations powered by Stable Diffusion (Diffusers)
- Panel-by-panel storyline and comic-style artwork
- Preview on the web, export as a structured PDF

"Your Imagination, Our AI!"



Key Features



AI

AI-powered story & image generation

Gemini + Stable Diffusion

P

Personalized comic creation

Based entirely on user input

I

Panel-by-panel illustrations

Each panel paired with narration

L

Live preview on web interface

Review the comic before export

D

Download comic in PDF format

Structured, ready to share

A

API endpoints + easy frontend

Form-based and JSON API access



1

Comic Creation (Serious/ Adventure)

- User enters a prompt, e.g. “a brave fox exploring an enchanted forest.”
- Provides character name, setting, tone, art style (e.g., anime)
- Gemini 1.5 Pro + Stable Diffusion generate the story and images

2

Customizing Mood & Style

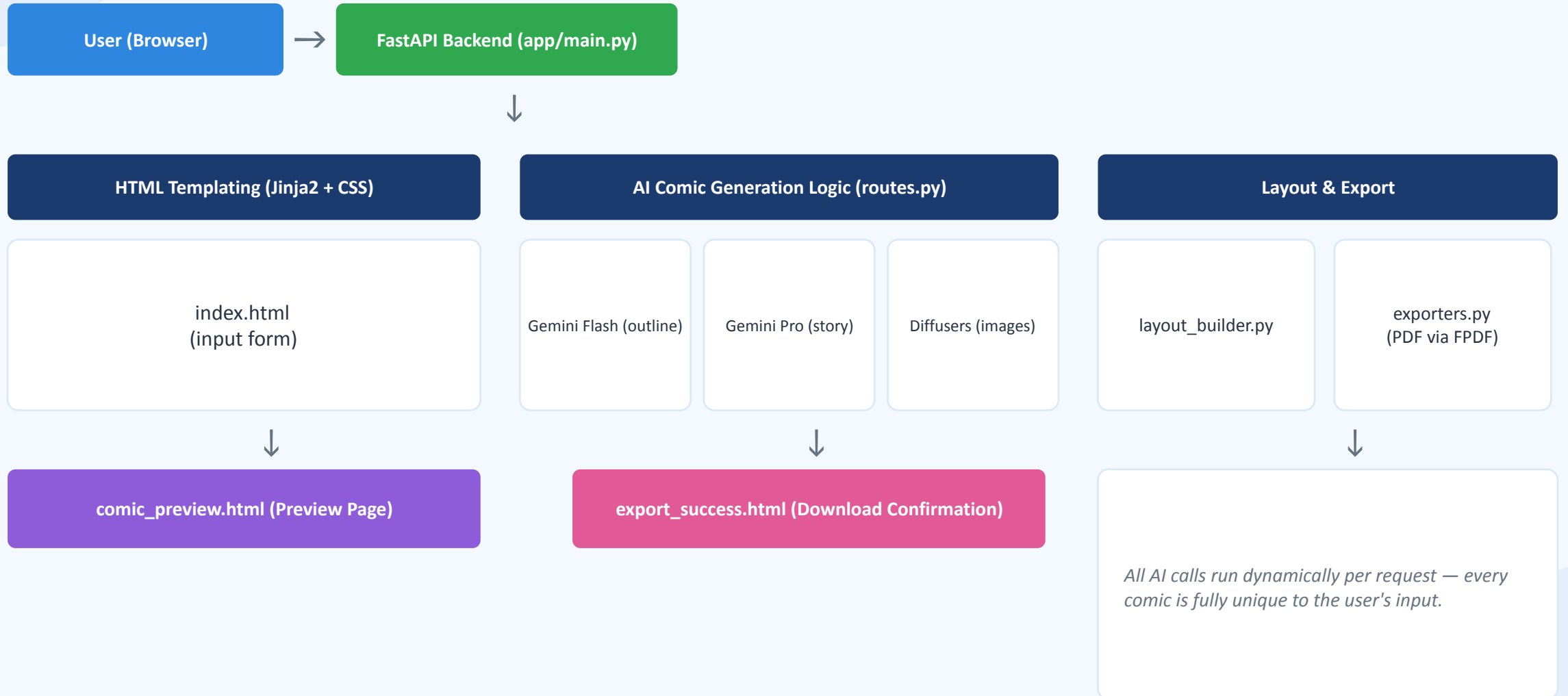
- User selects “funny” tone and “comic book” art style
- System regenerates the pipeline with tailored prompts
- Allows iteration until the comic matches their vision

3

Download & Export

- Layout binding via `layout_builder.py`
- PDF generation via `exporters.py` + FPDF
- Redirect to success page with download confirmation

Technical Architecture



Workflow



1

User Input

Story prompt, character name, setting, tone, art style

2

Story Generation (Gemini 1.5 Pro)

Comic outline, full story, dialogues

3

Image Generation (Stable Diffusion)

Panel-wise images in the chosen comic style

4

Layout Builder (layout_builder.py)

Combines image + text into a structured layout

5

PDF Export (exporters.py)

Generates a timestamped PDF using FPDF

6

Preview & Download

View the comic preview, then download the PDF

Technology Stack



1 Backend Framework

FastAPI

2 ASGI Server

Uvicorn

3 AI Models

Gemini 1.5 Pro, Gemini Flash

4 Image Generation

Stable Diffusion (Hugging Face Diffusers)

5 Frontend

HTML, CSS, Jinja2

6 PDF Library

FPDF

7 Version Control

Git

8 Language

Python

Pre-requisites

1 FastAPI Framework Knowledge

2 Uvicorn ASGI Server

3 Python Programming Proficiency

4 HTML, CSS and Jinja2 Template Skills

5 Google Gemini API Familiarity

6 Hugging Face Diffusers & Stable Diffusion

7 PyTorch Basics

8 PDF Generation with FPDF

9 Version Control with Git

10 Python Package & Environment Setup



Ready to build ComicCraft

Project Workflow



1

Milestone 1: Model Selection and Architecture

Research & select AI model; define architecture; set up dev environment

2

Milestone 2: Core Functionalities Development

Develop core functions; implement FastAPI routing & input processing

3

Milestone 3: routes.py Development

Write the main application logic in routes.py

4

Milestone 4: Frontend Development

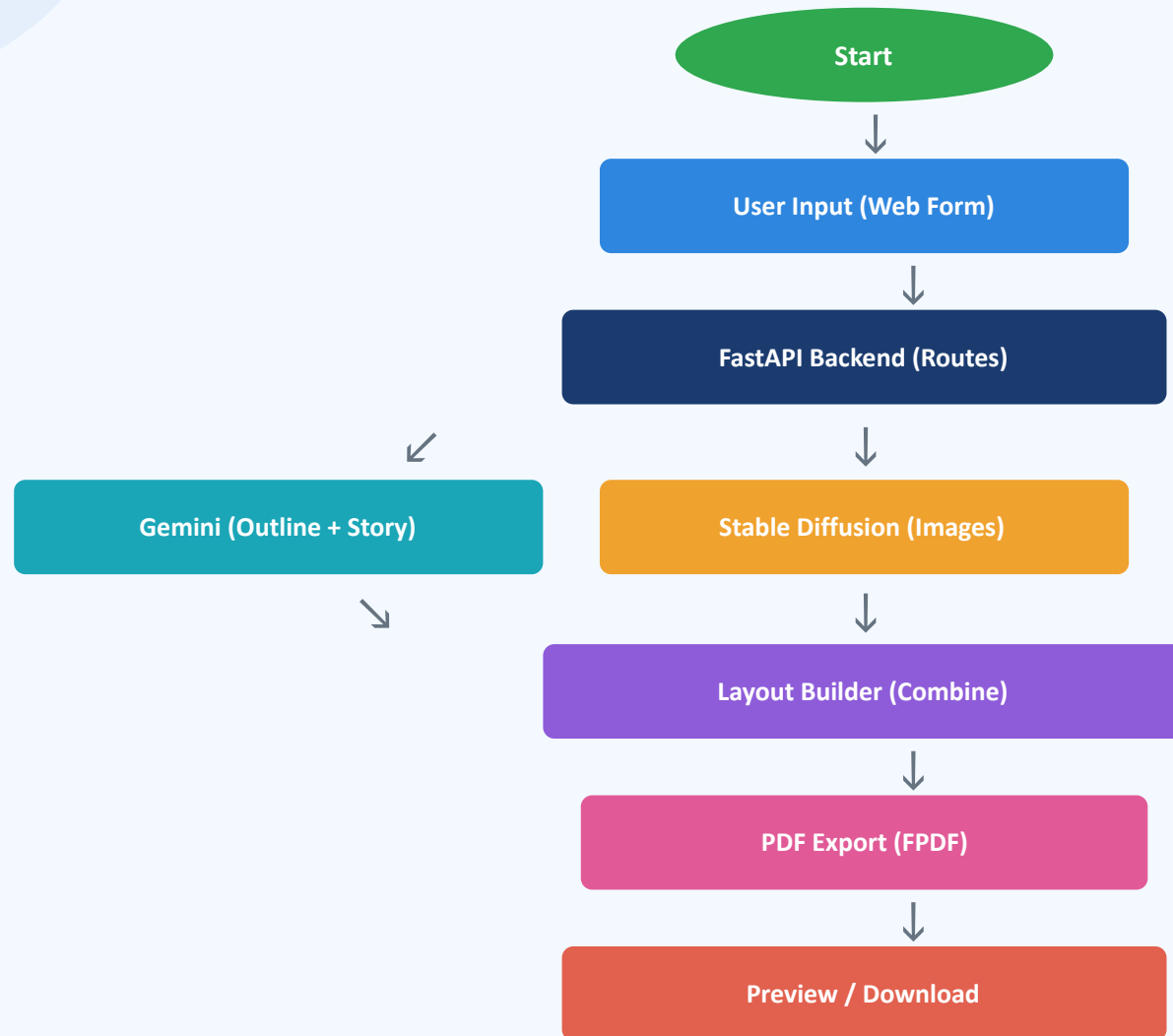
Design UI; create dynamic templates with Jinja2

5

Milestone 5: Deployment

Prepare and verify local deployment

Project Flow Diagram



Conclusion



ComicCraft makes comic creation easy, fast and accessible for everyone — even non-artists.



Combines the power of Gemini AI and Stable Diffusion for creative, personalized storytelling.

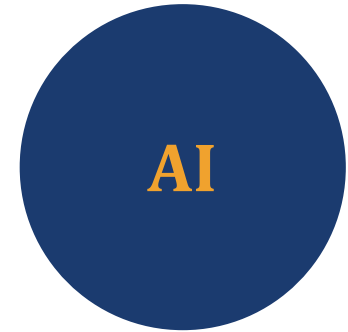


Built with FastAPI for a smooth and efficient backend.



Provides preview and PDF download for sharing, printing and saving.

Turn Your Ideas Into Comics!



A perfect blend of creativity + AI + modern web technologies.

Thank You!

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